Lesson 4

micro:bit Course

Module: Robotics

How to make Frankenstein



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MINT KidZ Schule

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Materials used



- Scissors
- Vinyl glue (PVA)
- Sticky tape
- Colours (black, green and grey)
- Black marker pen with a broad tip
- Paper clips
- 1 Micro:bit
- 1 180-degree servo motor
- 1 ELECFREAKS Micro:bit Ring:bit V2 Expansion Board
- Cable to connect the Micro:bit to the computer

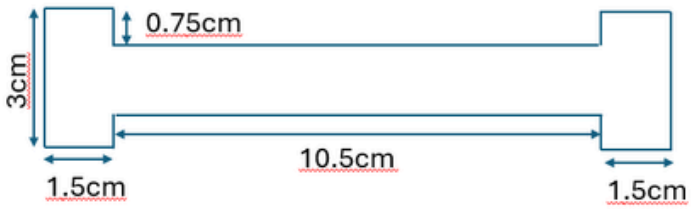
Cut the paper to these dimensions

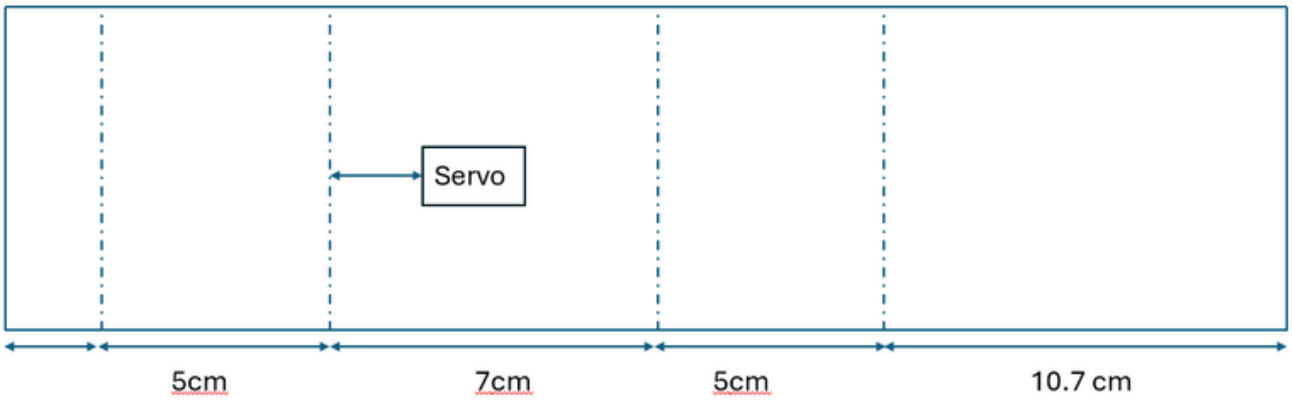
IMPORTANT

Download the net from the provided file in the course

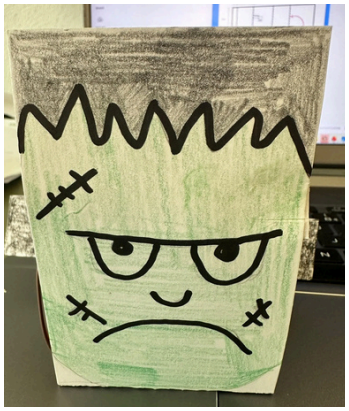
The net of the Frankenstein body

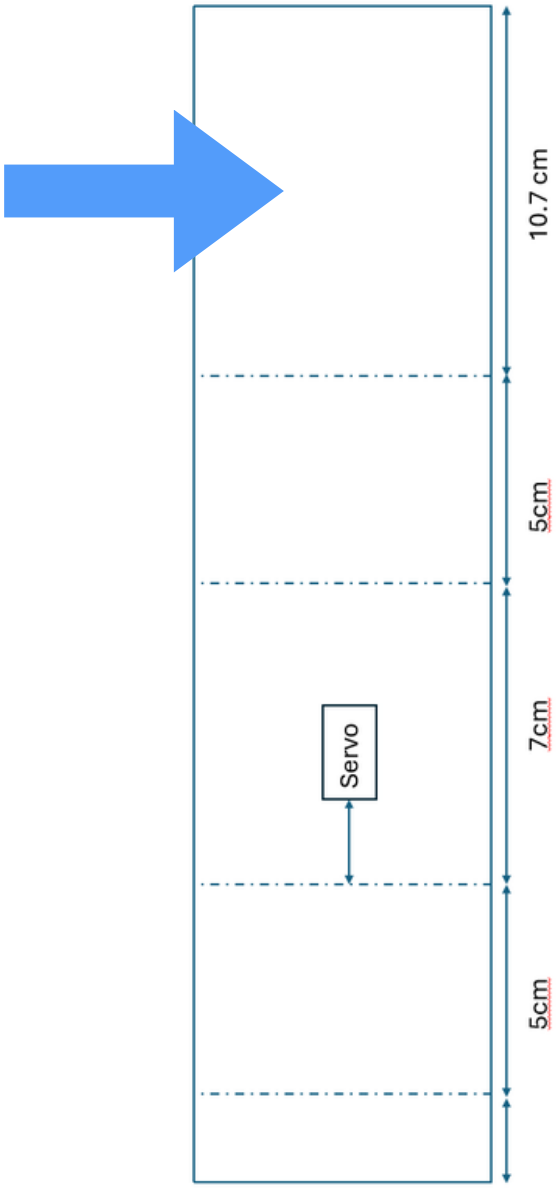
Cut the net along the solid line and fold it (using a ruler) along the dotted line.





Color your robots

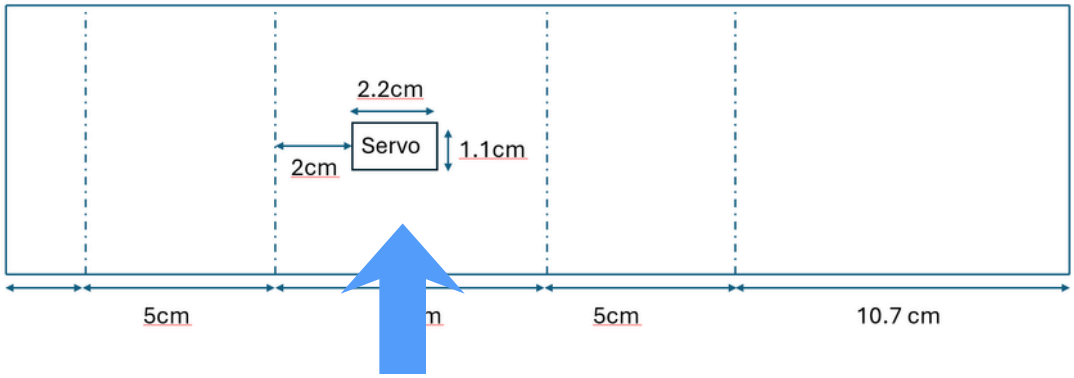




fColor this area with your Frankestei's face. Be creative! `

Cut

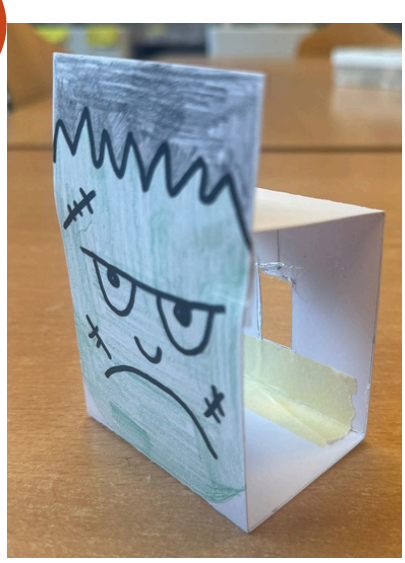
Cut out the space for the servo motor as shown in the figure

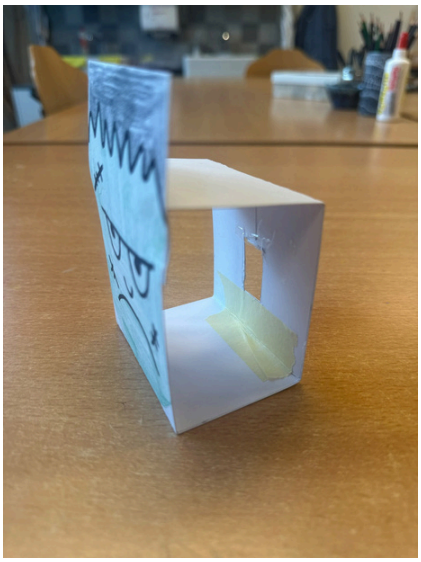


Asembly

Assemble the body of your robot as shown in the figure

1

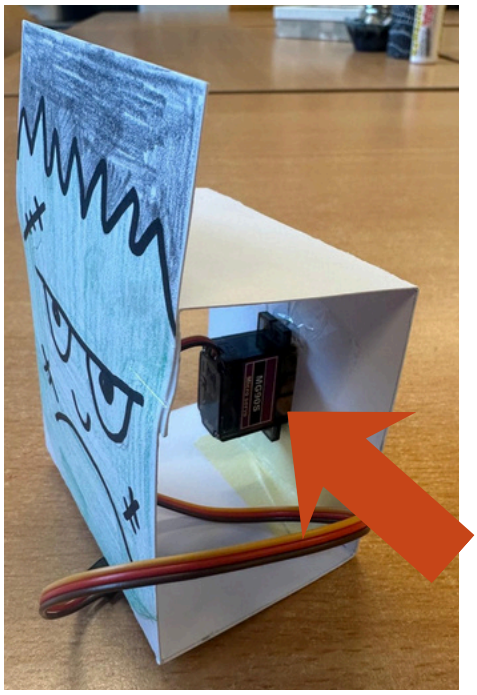




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Servo

2

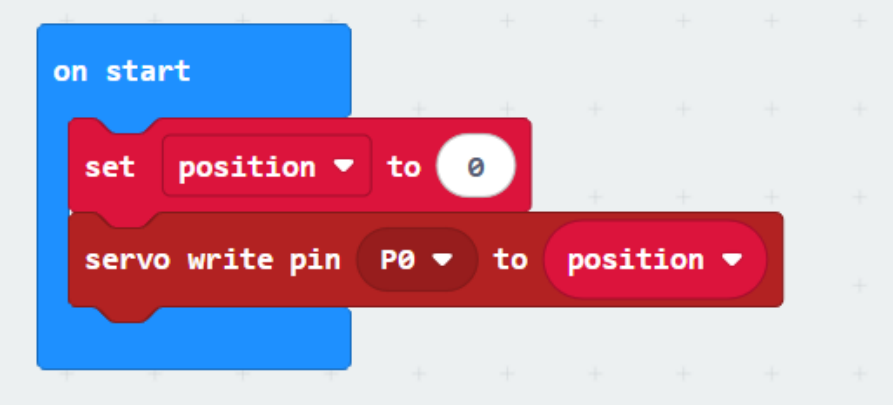


Place the servo in the hole and hold it in place with tape
Connect it to the Micro:bit

IMPORTANT

PInstructions on how to connect the Micro:bit to the servo motor can be found in Module 2

Position



How to program the servo motor was discussed in Module 2

Servo arm

3

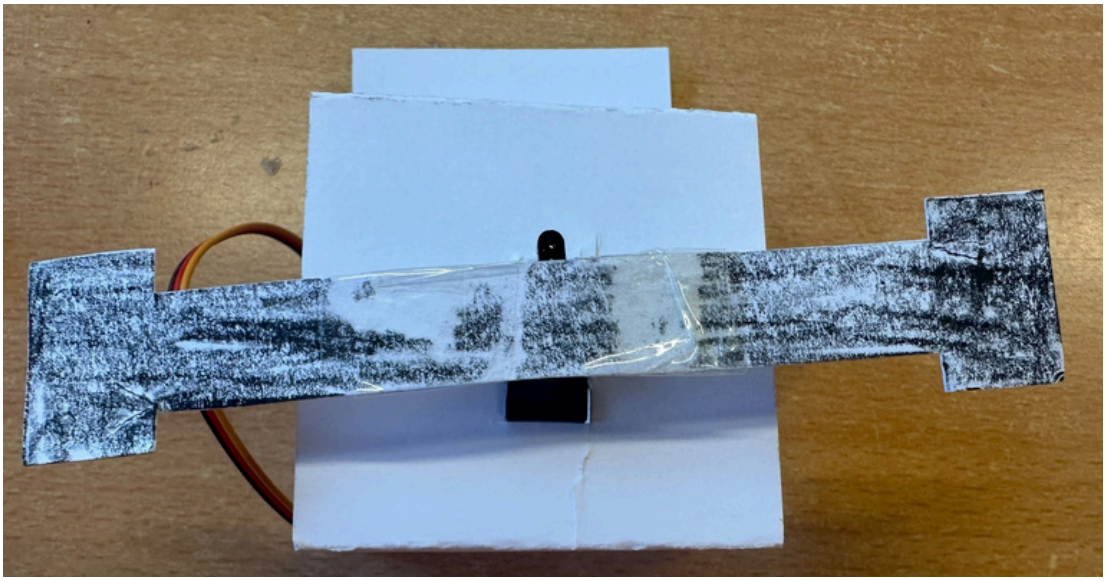
4



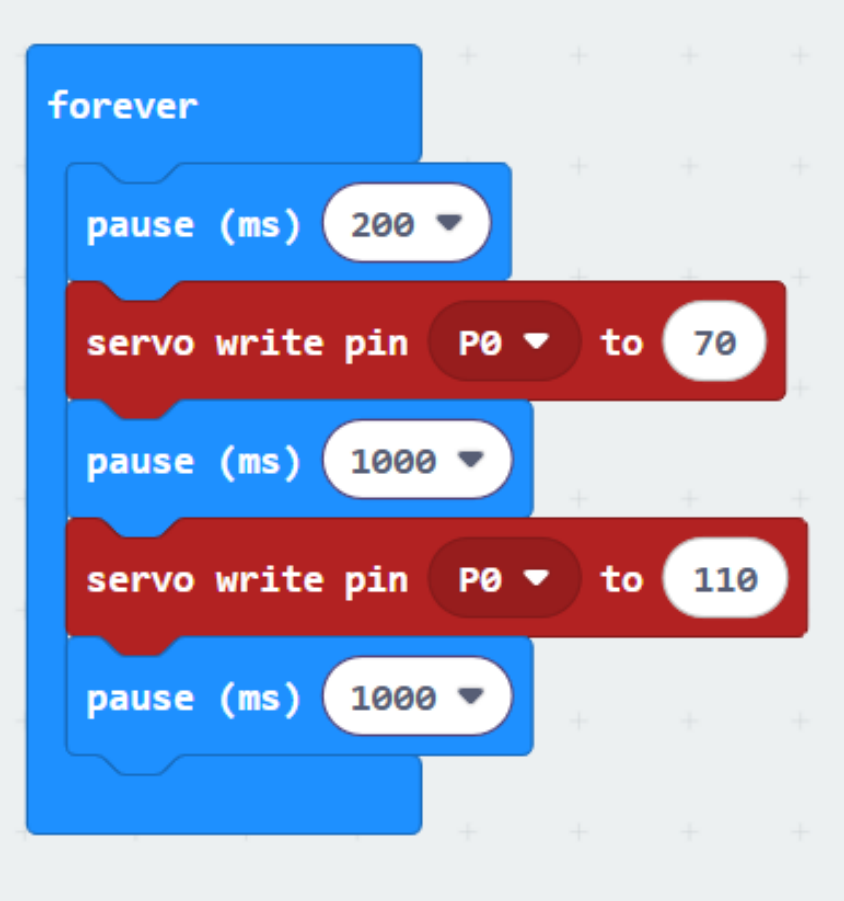
Connect the arm to the servomotor so that it is parallel to the table

5

Secure the “bolt” with a bit of tape



Coding


forever
pause (ms) 200
servo write pin P0 to 70
pause (ms) 1000
servo write pin P0 to 110
pause (ms) 1000



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